

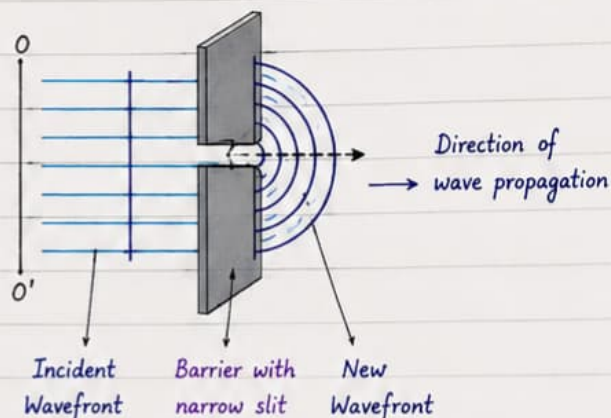
CHAPTER - 10

WAVE OPTICS

1. Wave Nature of Light

Light shows all the characteristics of waves such as interference, diffraction and polarization.

According to **Huygens' principle**, every point on a wavefront acts as a source of secondary wavelets. The new wavefront is the tangent drawn to these wavelets.

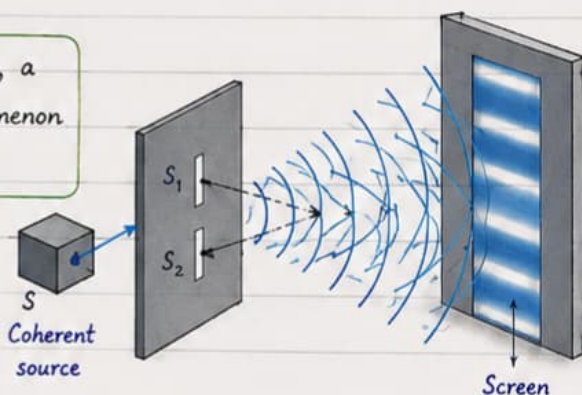


2. Interference of Light

When two or more coherent light waves superpose, a redistribution of intensity takes place. This phenomenon is called **interference**.

Conditions for sustained interference :

1. The sources must be **coherent** (same frequency and constant phase difference)
2. The sources must have same state of polarisation.
3. The sources must have same amplitude or nearly same.



Result of Interference :

Constructive Interference (Bright Fringe)

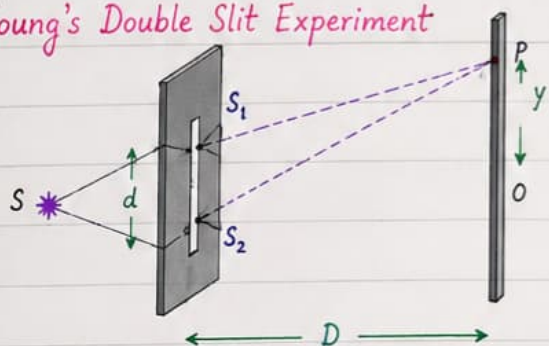
- Path difference (Δx) = $n\lambda$
($n = 0, \pm 1, \pm 2, \dots$)
- Intensity is maximum.

Destructive Interference (Dark Fringe)

- Path difference (Δx) = $(2n + 1) \frac{\lambda}{2}$
($n = 0, \pm 1, \pm 2, \dots$)
- Intensity is minimum (zero).

λ - Wavelength
 Δx = Path difference
 n = Order of fringe

Young's Double Slit Experiment



Fringe Width (β) :

$$\beta = \frac{\lambda D}{d}$$

Where,

λ = Wavelength of light
 D = Distance between slits and screen
 d = Distance between slits

Position of bright fringe :

$$y_n = n\beta$$

Position of dark fringe :

$$y_n = (2n + 1) \frac{\beta}{2}$$

($n = 0, \pm 1, \pm 2, \dots$)

Key Point Interference is the proof of the wave nature of light. ✨

3. Diffraction of Light

Diffraction is the bending and spreading of light around the edges of an obstacle or through a narrow opening.

Conditions for Diffraction to be noticeable :

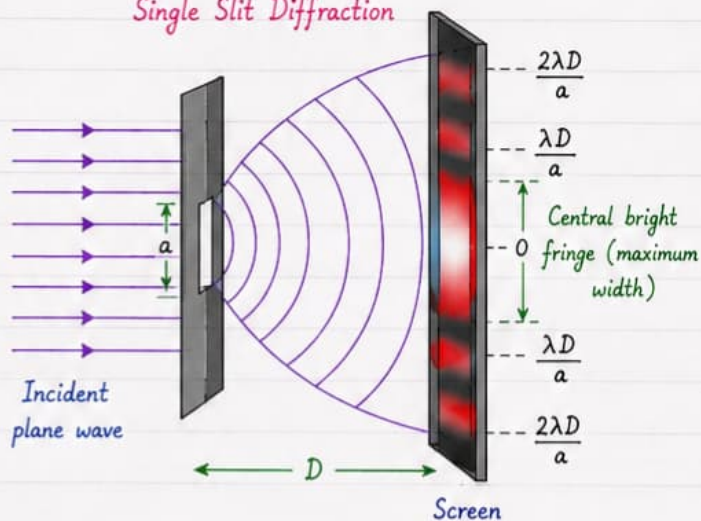
1. The size of aperture (a) should be of the same order as the wavelength (λ).
2. The screen should be placed far from the aperture.

Width of central bright fringe (β) :

$$\beta = \frac{2\lambda D}{a}$$

Where, λ = Wavelength of light
 a = Width of slit
 D = Distance of screen from slit

Single Slit Diffraction



Important Points

- ★ Central bright fringe is the widest and brightest.
- ★ Intensity of other fringes decreases rapidly.
- ★ Diffraction is maximum when slit is narrow.

4. Interference in Thin Films

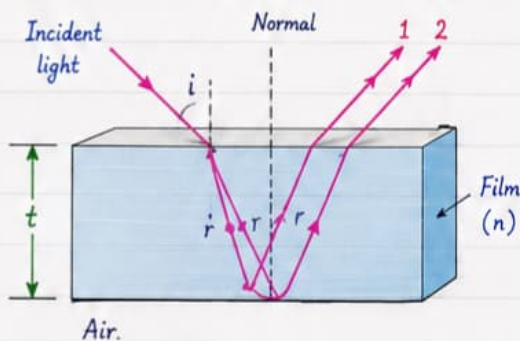
When light is incident on a thin transparent film (like soap bubble, oil film, water layer), part of the light is reflected from the top surface (1) and part from the bottom surface (2). These two reflected rays interfere.

Path Difference (Δ) :

For reflected rays,

$$\Delta = 2nt \cos r$$

Where, n = Refractive index of film
 t = Thickness of film
 r = Angle of refraction in film



Conditions for Constructive Interference (Bright Fringe)

When path difference is,

$$2nt \cos r = (2m + 1) \frac{\lambda}{2}$$

Where, $m = 0, 1, 2, \dots$

(One phase reversal occurs in reflected ray 2)

Conditions for Destructive Interference (Dark Fringe)

When path difference is,

$$2nt \cos r = m\lambda$$

Where, $m = 0, 1, 2, \dots$

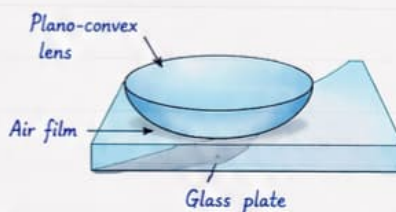
(No phase reversal or two reversals)

Examples of Thin Films

- Soap bubble
- Oil film on water
- Water layer on glass
- Newton's rings.

Newton's Rings

When a plano-convex lens is placed on a flat glass plate, a thin film of air is formed. On illuminating with monochromatic light, alternate bright and dark concentric rings are seen.



Radius of m^{th} dark ring :

$$r_m = \frac{\sqrt{m\lambda R}}{n}$$

Radius of m^{th} bright ring :

$$r_m = \sqrt{\left(m - \frac{1}{2}\right) \frac{\lambda R}{n}}$$

Where,

R = Radius of curvature of lens
 n = Refractive index of medium between lens and plate (for air, $n = 1$)
 λ = Wavelength of light

Key Point

Newton's rings are due to interference in a thin film of variable thickness.

Remember : Wave optics shows that light behaves as a wave in certain situations.

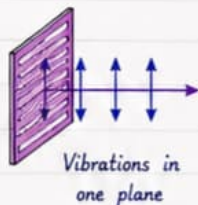
5. Polarization of Light

Polarization is the phenomenon in which the vibrations of light are restricted to one particular direction perpendicular to the direction of propagation.

Types of Polarization :

1. Plane Polarization :

- Vibrations occur in only one plane.
- Example: Light from a Polaroid.



2. Unpolarized Light :

- Vibrations occur in all possible planes perpendicular to the direction of propagation.
- Example: Sunlight, bulb light.



Malus' Law :

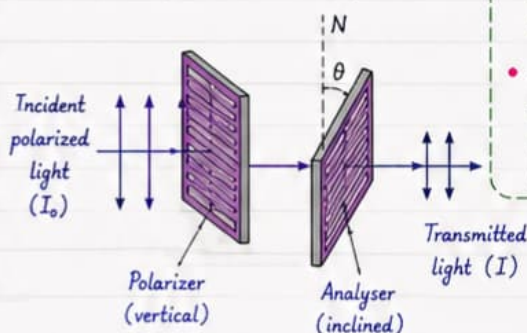
When plane polarized light of intensity I_0 is incident on an analyser and the axis of analyser is inclined at an angle θ with the direction of vibration, the intensity of transmitted light is

$$I = I_0 \cos^2 \theta$$

Where,

I_0 = Intensity of incident polarized light

I = Intensity of transmitted light

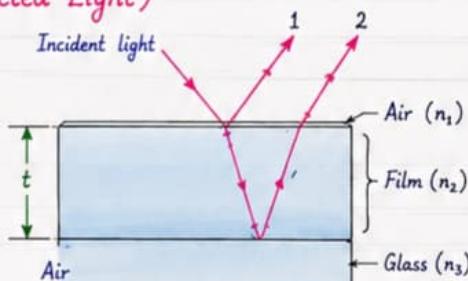


Key Point

- Maximum transmission when $\theta = 0^\circ$
- Minimum transmission when $\theta = 90^\circ$ ($I = 0$)

6. Interference in Thin Films (Reflected Light)

When a thin film is illuminated by light, reflections from the top and bottom surfaces interfere and produce bright and dark fringes.



Path difference,

$$\Delta = 2 \mu t \cos r$$

Where,

μ = Refractive index of film

t = Thickness of film

r = Angle of refraction in film

Conditions for Reflected Light

Bright Fringe (Constructive)

$$\Delta = (2m + 1) \frac{\lambda}{2}$$

or

$$2 \mu t \cos r = (2m + 1) \frac{\lambda}{2}$$

Where, $m = 0, 1, 2, \dots$

Dark Fringe (Destructive)

$$\Delta = m \lambda$$

or

$$2 \mu t \cos r = m \lambda$$

Where, $m = 0, 1, 2, \dots$

For normal incidence ($r = 0^\circ$)

$$\text{Bright : } 2 \mu t = (2m + 1) \frac{\lambda}{2}$$

$$\text{Dark : } 2 \mu t = m \lambda$$

Examples

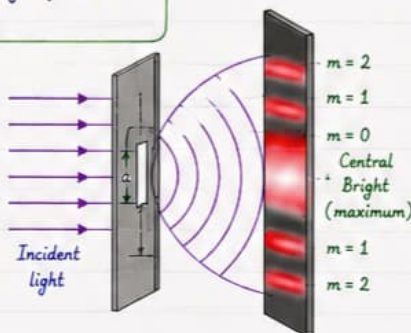
- Oil film on water
- Soap bubble
- Newton's rings

7. Diffraction of Light

Diffraction is the bending of light around the edges of an obstacle or through a narrow opening of size comparable to the wavelength of light.

Single Slit Diffraction

A single slit of width a produces a central bright fringe surrounded by alternate dark and bright fringes of decreasing intensity.



Conditions for Minima (Dark Fringes)

$$a \sin \theta = m \lambda$$

Where, a = Width of slit

θ = Angle of direction

$m = \pm 1, \pm 2, \pm 3, \dots$

Conditions for Maxima (Bright Fringes)

Approximately at

$$a \sin \theta = \left(m + \frac{1}{2}\right) \lambda$$

Where, $m = 0, \pm 1, \pm 2, \dots$

Key Point

With increase in slit width (a) $\rightarrow$ fringes become narrower.
With increase in wavelength (λ) $\rightarrow$ fringes become wider.

Remember : Wave optics explains the wave nature of light in phenomena like interference, diffraction and polarization.

8. Diffraction Grating

A diffraction grating consists of a large number of parallel, equally spaced slits.

Principal maxima (brightness):

$$d \sin \theta_m = m \lambda$$

$$(m = 0, \pm 1, \pm 2, \pm 3 \dots)$$

Where,

d = Grating element (distance between two consecutive slits)

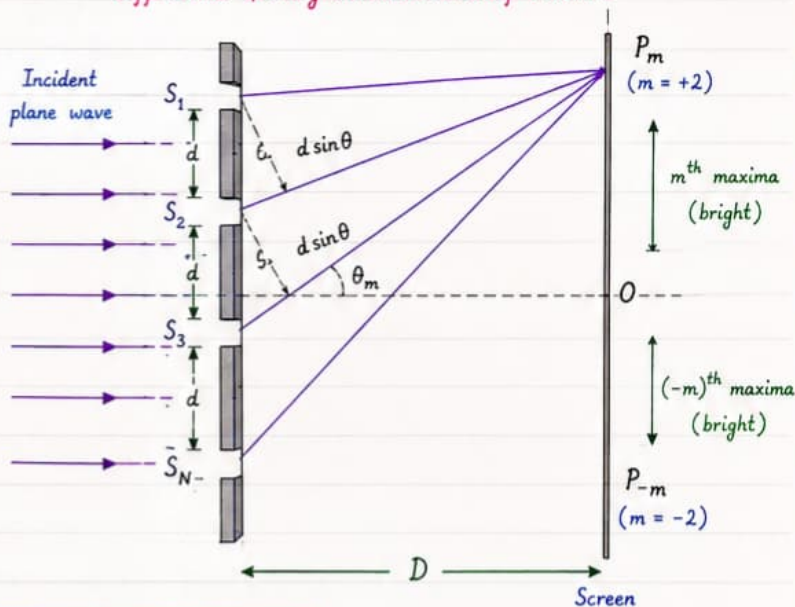
θ_m = Angle of m^{th} principal maxima

λ = Wavelength of light

m = Order of maxima

★ Central maxima ($m = 0$) occurs at $\theta = 0^\circ$.

Diffraction Grating and Formation of Maxima



Properties of Diffraction Grating

- ★ It produces a large number of sharp and intense maxima.
- ★ Angular separation between successive maxima is constant.
- ★ Resolving power of grating is very high.

- ★ Dispersive power is higher than a prism.
- ★ Used in spectrometers to obtain high resolution spectra.



9. Resolving Power of Grating

It is the ability of a grating to resolve two close wavelengths (λ and $\lambda + \Delta\lambda$).

Formula :

$$R = \frac{\lambda}{\Delta\lambda} = mN$$

Where,

R = Resolving power

λ = Wavelength

$\Delta\lambda$ = Smallest difference in wavelength that can be resolved

m = Order of maxima

N = Total number of slits illuminated

Key Point

Resolving power increases with increase in order (m) and number of slits (N).



Comparison of Interference and Diffraction

Basis	Interference	Diffraction
Cause	Superposition of waves from two or more coherent sources.	Bending of light around edges or through narrow openings.
Fringe Width	Nearly equal and constant.	Central maximum is widest; others have unequal width.
Intensity of Fringes	Bright and dark fringes have nearly equal intensity.	Intensity decreases rapidly away from central maximum.
Dependence	Depends on path difference.	Depends on aperture size and wavelength.
Example	Young's double slit experiment.	Single slit diffraction and grating.

10. Applications of Wave Optics

- Determination of wavelength of light (using interference and diffraction).
- Measuring very small distances.
- Spectrometers for studying spectra of stars and elements.
- Testing the flatness of optical surfaces (interference).
- Anti-reflection coatings on lenses.
- Holography, optical fibers and many more.



Anti-reflection coating



Optical fiber



Holography

Quick Revision

□ Interference → path difference (Δ)

Bright : $2n\lambda$
Dark : $(2n+1)\frac{\lambda}{2}$

□ Diffraction (single slit) → minima at $a \sin \theta = n \lambda$

Central maxima width $\beta = \frac{2\lambda D}{a}$

□ Thin films → Path difference $\Delta = 2\mu t \cos r$

Bright : $2\mu t \cos r = (2m+1)\frac{\lambda}{2}$
Dark : $2\mu t \cos r = m\lambda$

□ Grating → Principal maxima : $d \sin \theta_m = m \lambda$

Resolving power : $R = \frac{\lambda}{\Delta\lambda} = mN$

□ Polarization → $I = I_0 \cos^2 \theta$

Summary

Wave optics reveals the wave nature of light through interference, diffraction and polarization. These phenomena have wide applications in modern optical instruments.

Study Smart
Score Higher!

